

ALTMAN Z-SCORE: OVERVIEW

Z-SCORE (ALTMAN 1968)

- A composite score used to predict corporate failure (ie, *bankruptcy probability*) up to 2 years:
 - $$\text{Z-score} = a_1 (\text{CA-CL})/\text{TA} + a_2 \text{RE}/\text{TA} + a_3 \text{EBIT}/\text{TA} \\ + a_4 (\text{MV of Stock})/(\text{BV of TL}) + a_5 \text{Sales}/\text{TA}$$
 - with coefficients: $a_1 = 1.2$, $a_2 = 1.4$, $a_3 = 3.3$,
 $a_4 = 0.6$, and $a_5 = 1.0$
 - CA = Current assets; CL = Current liabilities; TA = Total assets
RE = Retained earnings; EBIT = Earnings before interest and taxes
MV of Stock = “combined market value of all shares of stock,
preferred and common”
BV = Book value; TL = Total liabilities
- scores < 1.81 for high-risk firms and > 2.99 for low-risk;
in-between for unclear indication

COMPONENT RATIOS OF Z-SCORE

- Because $a_i > 0$,
 - a higher component ratio raises the Z-score
- **Sales/TA**: simply the *asset turnover*
 - Should averaging with prior year be applied on TA before the ratio computation? No
 - To be consistent with Altman (1968) so that *cutoffs at 1.81 and 2.99 are valid*
 - Similarly for all the component ratios
- **(CA-CL)/TA**: a *liquidity measure* like current ratio (CA/CL)
 - Holding constant either CA or CL,
 - larger (CA-CL) difference means higher CA/CL ratio

COMPONENT RATIOS OF Z-SCORE

- **RE/TA** may be viewed as a *cumulative* version of return on assets (ROA = NP/TA)
 - Retained Earnings (RE) accumulates all the net profits (NPs) of the current and past years – if there has not been any dividend distribution
- **EBIT/TA**: a *profitability* ratio like ROA = NP/TA
 - can be expressed in other ratios already discussed: $\frac{EBIT}{Sales} \times \frac{Sales}{TA}$
- **(MV of Stock)/(BV of TL)**: a *solvency measure* like the inverse of debt-to-equity ratio (Total equity/Total debt)
 - measures how far MV of TA can fall, as a multiple of BV of TL, before a company becomes insolvent
 - a company becomes *insolvent* when MV of TA is no longer at least BV of TL – more likely when MV of Stock drops down to 0

Z-SCORE - EXAMPLE

Selected financials and stock info of Oshkosh Corporation from its website for FY ended 30/90/2019:

(in USD'000, except share price)	FY 2019	(in USD'000, except share mounts)	30/09/2019
Net sales	8,382,000	Cash and cash equivalents	448,400
Cost of sales	6,864,600	Total current assets	3,408,300
Gross profit	1,517,400	Gross property, plant and equipment	1,360,900
Selling general and administrative	683,500	Accumulated depreciation	(787,300)
Total operating expenses	720,400	Total non-current assets	2,158,000
Operating income	797,000	Total assets	5,566,300
Interest expense	54,400	Current debt, incl. curr. maturities of LT debt	-
Interest income	6,800	Taxes payable	73,500
Income before tax	750,700	Total current liabilities	1,741,900
Provision for income taxes	171,300	Long-term debt, less curr. maturities	819,000
Net income	579,400	Other long-term liabilities	405,600
Dividends paid	75,500	Total liabilities	2,966,500
Cash paid for interest	53,600	Preferred Stock (\$.01 par value; 2,000,000 shares authorized; none issued & outstanding)	-
Cash paid for income taxes	117,600	Common stock (\$.01 par value; 300,000,000 shares authorized; 75,101,465 shares issued)	700
Depreciation and amortization	115,200	Additional paid-in capital	808,500
Capital expenditures	174,200	Retained earnings	2,505,000
		Common Stock in treasury, at cost (7,114,349 shares)	(512,800)
Closing price of common stock (30/09/2019)	75.8	Total stockholders' equity	2,599,800

Z-SCORE - EXAMPLE

Q: What is the company's Z-score for the year?

(All figures below in USD'000, *except* ratios, share price, and share amounts)

- $(CA-CL)/TA = (3,408,300 - 1,741,900)/5,566,300 = 0.30$
 $RE/TA = 2,505,000/5,566,300 = 0.45$
- $EBIT = EBT + (\text{Interest expense} - \text{Int. income}) = 750,700 + (54,400 - 6,800)$
 $EBIT/TA = 798,300/5,566,300 = 0.14$
- Shares outstanding
= Shares issued - Shares in treasury = 75,101,465 - 7,114,349
 - Since there's *no preferred shares outstanding*,
MV of Stock = (\$75.8 per share × 67,987,116 common shares)/1,000
 - $(MV \text{ of Stock})/(BV \text{ of TL}) = 5,153,423/2,966,500 = 1.74$
- $Sales/TA = 8,382,000/5,566,300 = 1.51$
- $Z\text{-score} = 1.2(0.30) + 1.4(0.45) + 3.3(0.14) + 0.6(1.74) + 1.0(1.51) = 4.01$
 - has a *low risk* of bankruptcy (< 1.81 for *high-risk* and > 2.99 for *low-risk*)

CREDIT ANALYSIS

- Altman Z-score was originally developed to determine which financial ratios are useful in assessing the risk of bankruptcy
 - therefore is often used for **credit analysis**
- Credit analysis refers to the evaluation of **credit risk**
 - i.e., the risk of loss caused by a counterparty's or debtor's *failure to make a promised payment*
 - eg, a bond issuer not being able to pay interest and principal as due

CREDIT RATING

- A **credit rating** is an indication of the opinion expressed by a rating agency
 - after the process to assess an entity's credit risk or *overall creditworthiness* (ie, how worthy it is to be granted credit)
- The rating concerns
 - the *probability of default* by an issuer on its debt obligations (eg, notes and bonds)
 - or, an entity's *overall capacity and willingness* to meet its financial obligations, if it has no debt outstanding
- Part of the process involves using some financial ratios, referred to as **credit ratios**, to analyze an entity's *cash flow, leverage, and competitive position*